

YOUSEF ALOGIELY

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SUMMARY

Fresh Computer Science graduate specializing in AI and backend development, building machine learning systems for real-world applications. Focused on developing impactful intelligent systems and advancing the field.

EDUCATION

Bachelor of Computer Science

September 2021 – June 2026

Qassim University, College of Computer, Buraydah

GPA: 4.97 out of 5.00 | Class Rank: 1st among graduating students

Artificial Intelligence Programme

July – August 2026

University of Oxford, Lady Margaret Hall (LMH)

- Attended the **University of Oxford AI Summer Programme** at Lady Margaret Hall with a highly competitive international cohort, across three intensive stages: deep learning, NLP, and agentic AI.
- Deep learning: implemented Transformers, Vision Transformers, GANs, VAEs, Diffusion Models, and Graph ML; built Batch Normalization from scratch and fine-tuned ViT and Swin Transformers for medical image classification with a sliding-window pipeline for high-resolution images.
- NLP: built and evaluated text pipelines spanning tokenization, TF-IDF, Word2Vec and GloVe embeddings, RNNs, LSTMs, GRUs, CNNs, and transformer-based models.
- Agentic AI: built LLM agents with tool calling and API integration, memory and multi-step planning, RAG over vector search, and multi-agent orchestration in Python, evaluated through run tracing, guardrails, and cost control.

EXPERIENCE

AI Trainee — KAUST Academy

January – September 2026

King Abdullah University of Science and Technology (KAUST), Thuwal

- Studied and applied machine learning, deep learning, and computer vision on object detection and image segmentation tasks. Applied transfer learning, fine-tuning, and data techniques to improve generalization, stability, and performance, reflecting strong capability in building, evaluating and understanding deep learning systems.
- Selected among the top 1% of more than 24,000 applicants for the KAUST Academy Summer Program, earning a place in a highly selective AI training track.

Maintenance Specialist

August 2023 – December 2023

Deanship of E-Learning and Information Technology, Qassim University

- Collaborated within a team to improve core system performance and structure, reducing response times and increasing maintainability, while identifying and resolving system issues to ensure stable operation across high-traffic university systems.

PROJECTS

Watheeq – AI Cybersecurity Compliance Platform (Graduation Project) | 2026

Technologies: Python, Django, React, LangChain, LLaMA-4, RAG

[GitHub](#) | [Live Demo](#)

- Led a 4-person team to build an AI system mapping user inputs to relevant cybersecurity controls across NCA, SAMA, and ECC frameworks.
- Developed a RAG pipeline using LangChain, achieving 94% retrieval accuracy across all three frameworks.
- Built a Django backend with PostgreSQL and JWT authentication for secure concurrent multi-user access.

RR Layer – SVD-Based Batch Normalization (KAUST Academy Research Project) | 2026

Technologies: Python, PyTorch, NumPy, CUDA, Matplotlib

[Project Page](#) | [Live Demo](#)

- Implemented the Rank Reduction (RR) layer as a standalone PyTorch module: a truncated-SVD projection placed where BatchNorm sits, between a convolution and its activation.
- Benchmarked plain, BatchNorm, and SVD-layer networks from identical initial weights over 32 seeds: 3.9–15x lower test MSE than BatchNorm and 1.6–3.0x lower than the plain network at every training-set size under heavy input noise.
- Ran rank, noise, dataset-size, and background ablations, and published an interactive research site with a live demo.